

Control Systems (EEE/ECE/INSTR F242)

Tutorial 8 (3 March 2026)

- For the system shown in Fig. Q1, find N_1/N_2 so that the settling time (2% tolerance band) for a step torque input is 16 seconds. Also, find the value of damping ratio, natural frequency and static gain of the system for this value of N_1/N_2 .

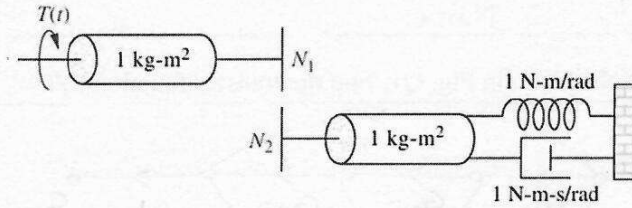


Fig. Q1

- A motor and generator are set up to drive a load as shown in Fig. Q2. If the generator output voltage is $e_g(t) = K_f i_f(t)$, where $i_f(t)$ is the generator's field current.
 - Find the transfer function $G(s) = \theta_o(s)/E_i(s)$ in zero-pole-gain form. For the generator, $K_f = 2\Omega$. For the motor, $K_t = 1 \text{ N-m/A}$, and $K_b = 1 \text{ V-s/rad}$.
 - Assume, that the θ_o is to be controlled using a suitable feedback control loop with a proportional controller having gain K_c . Find the value of K_c to have a steady state error of 0.01 rad for a ramp reference input. Assume feedback gain to be unity.

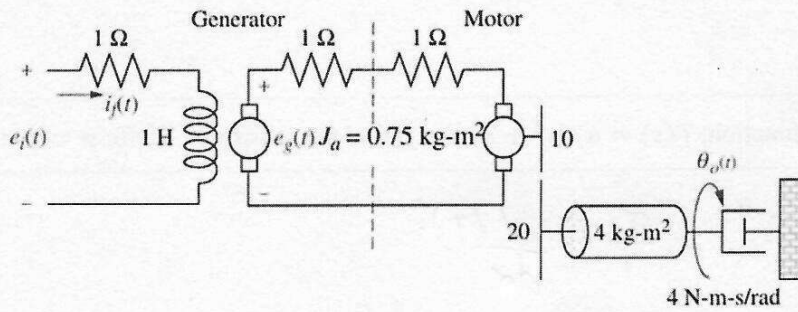
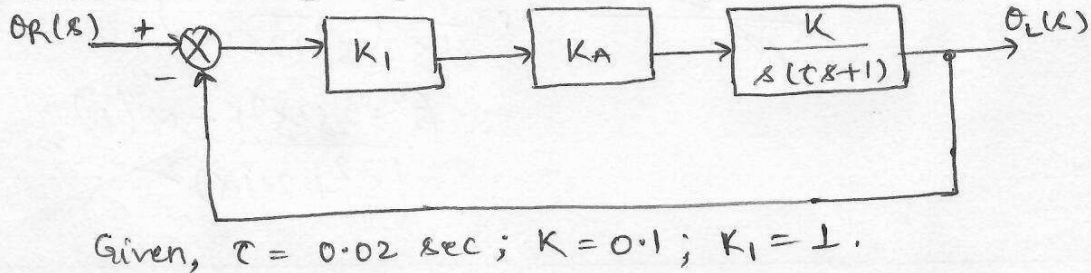


Fig. Q2

- Consider a schematic diagram of position servomechanism system shown in Fig. Q3.



- Find the value of ω_n and ζ for an amplifier gain of $K_A = 500 \text{ V/V}$. What is the steady state error for this value of K_A for a unit step and a unit ramp input.
- Assume a PD controller is now also used having a transfer function of $\left(\frac{K_d}{K_A}s + 1\right)$. Find the value of K_d such that closed loop dynamics have a relative damping ratio of unity.
